Evaluating Intertwined Effects in Fake News Spreading by using MCDM Approaches

Bo Hsiao
Department of Information Management
National Yunlin University of Science and Technology
Yunlin, Taiwan
bhsiao@yuntech.edu.tw

***Abstract*—What news is real and fake has formed a hot topic in recent years. The current discussions concerning fake news indicate that academics and practices still lack consensus on measuring the influencing factors. For dealing with this issue, the purpose of this study is to focus on investigating and assessing influence factors to reduce the gaps in spreading of fake news by interdependence and feedback problems among dimensions and criteria to find the most significant factors. We used DEMATEL-based Analytic Network Process (DANP) to solve these problems. Then we conducted several experts to investigate the root-and-cause of spreading of fake news. These results can provide government understanding of how to prevent strategies and fill the gaps between academic and practice.**

Keywords*—*fake news, multiple criteria decision-making, DANP

I. INTRODUCTION

Allcott and Gentzkow[1] defined fake news as “news articles that are intentionally and verifiably false and could mislead readers.” Tong et al. [2] supported this argument by claiming that “fake news took off simultaneously, and news began to circulate widely.” Fake news usually refers to news articles that are deliberately and provably false and thus may mislead readers. When people spread false fake news, they usually have a specific purpose or intention. For example, they spread fake news to persuade people to support or oppose individuals, groups, ideas, goals, thoughts, or future actions, or criminal activities. Apart from the above reasons, fake news could also exaggerate a situation. People can use them to confound past events and activities, or explain the significance of discovering false information.

Although scholars have made progress in discovering incorrect information and fake news in recent years, the process remains challenging due to the complexity, diversity, multimodality, and cost (e.g., checking or annotation) of fake news or false information[3]. Consequently, the widespread dissemination of incorrect information has a severe negative effect on individuals and society. Fake news also affects a person’s confidence in the news ecosystem.

Who will influence and be influenced by the fake news and how to spread the fake news becomes an important issue in this age. Social psychology points out that social influence (i.e., the spread of news articles) and self-influence (i.e., users’ existing knowledge) are effective agents for spreading false information. The more significant effect from society and oneself can distort users’ perceptions and behaviors, making them believe in a news article and unintentionally participate in its dissemination. Besides, disseminating false or misleading information is often dynamic, leading to disorderly exchanges between different information types [4]. For example, fraudulent information producers can deliberately spread false news on social platforms. People who see this information may not know that it is fake and then use their relationship to share this information in their communities.

Social media’s openness and anonymity make use convenient to share and exchange information, making them vulnerable to illegal activities. These situations also help the fake news spreading to grow up and made the information disorder. Although scholars studied phonemes, the openness of such platforms and automation’s potential have caused the disorderly and rapid spread of information to many people, which has brought unprecedented challenges. For figuring and casue and effect of fake news spreading, information disorder has been appeared a critical issue and attracted much attention in recent years [5]. Literature reviews

This section aims to analyze the cause-and-effect of influencing factors of fake news pass through the five facts (i.e., “technological environment,” “content characteristics,” “motivation factors,” “social-psychological behaviors,” and “personality characteristics”) to simplify the research problem and understand the whole picture. Following the previous discussion, we can summary fake news features into different factors as shown in Figure 1.

II. METHODOLOGY

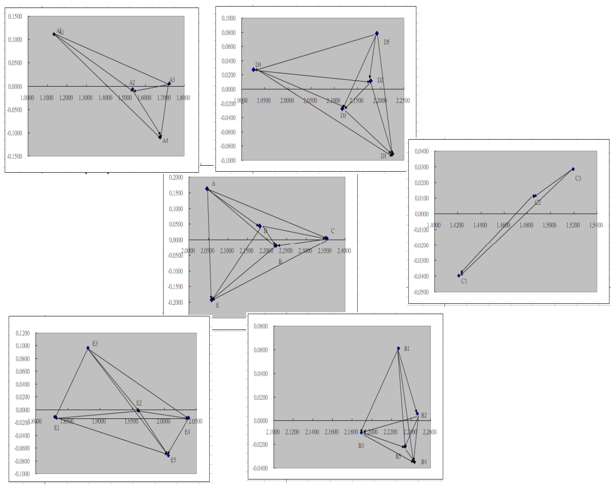
Before we talking about the cause-and-effect of fake news, this project would like to brief introduce the basic concept of DANP for unfamiliar readers. The detail steps could be referred to Tzeng et al.[6])

III. RESULTS

As previous discussed, the criteria and indicators from various studies, 5 main criteria and 22 subcriteria were defined to select fake news spreading. This study has several findings. First, in the INRM, this result presents an easily big map from Figure 1 and Table 1 that the five dimensions influences each other. For example, A (i.e., Technological environment) affects dimensions B (i.e., Content Characteristics), C (i.e., Motivation Factors), D (i.e., Social-Psychological Behaviors), and E (i.e., Personality Characteristics); dimension B affects dimension E; dimension C affects dimensions B and E; dimension D affects dimensions B, C, and E, Understanding these influential relationships will enable decision-maker to know which factors will affect the spreading of fake news.

Next finding was influence weights. Table 1 shows that the highest relative weights were C2 (Ideology) and C3 (Manipulating Public Opinion), assessed at 0.077 and 0.077, respectively. This result indicates that spreading fake news and “manipulation” of fake news has the greatest relative influence weights. Similarly, we also demonstrate the importance of influence on dimensions. This appears to be an exciting finding that the Motivation Factors are the most important from influence viewpoints, followed the Social-Psychological Behaviors, technological environment, Content Characteristics, and personal characteristics.

FIG1. INFLUENCE NETWORK MAP



Thus, it is essential to prevent the manipulation of public opinion and ideology. Figure 1 and Table 1 illustrate the priority of criterion removal from top to bottom, removing the most influential criteria for reducing gaps and meeting stockholders' needs.

IV. CONCLUSION

Pass through DANP, this study calculated a weighted and unwanted super-matrix to determine the influential weights based on the total influential matrix. This study can help decision-makers discuss spreading fake news redesign—the influence weights questionnaire from experts.

ACKNOWLEDGMENT

This research is supported by National Science and Technology Council of Taiwan under project number NSTC 111-2410-H-224-038 and 110-2410-H-224 -039.

REFERENCES

[1] H. Allcott, and M. Gentzkow, "Social Media and Fake News in the 2016 Election," Journal of Economic Perspective,vol. 31(2). Pp211-236, 2017.

[2] C. Tong, H. Gill, J. Li, S. Valenzulea, and H. Rojas,"Fake News is anything they say!-Conceptualization and Weaponization of Fake news among American Public,", Mass Communication and Society, Vol. 23, pp. 755-778, 2020.

[3] M. Schudson, and B. Zelizer, (2017), "Fake news in context in understanding and addressing the disinformation ecosystem." Understanding and Addressing the Disinformation Ecosystem", 2017.

[4] C. Wardle, and H. Derakhshan,"Information disorder: Toward an inter- disciplinary framework for research and policy making" Council of Europe Report, 27.2017.

[5] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, "Language models are unsupervised multitask learners", OpenAI Blog,1(8), 2019.

[6] G.H. Tzeng, C.H. Chiang, & C.W. Li, (2007). Evaluating intertwined effects in e-learning programs: A novel hybrid MCDM model based on factor analysis and DEMATEL. Expert Systems with Applications, Vol. 32, pp. 1028–1044, 2007.

TABLE I. INFLUENCE WEIGHTS (LOCAL AND GLOBAL)

	d	r	$d+r$	$d-r$	Global Weight	Sub system	d	r	$d+r$	$d-r$	Local weight
A	1.105	0.942	2.046	0.163	0.204	**A1**	0.511	0.622	1.133	0.111	0.036
						A2	0.772	0.765	1.537	-0.007	0.055
						A3	0.860	0.865	1.725	0.005	0.061
						A4	0.893	0.783	1.675	-0.110	0.063
B	1.102	1.121	2.223	-0.020	0.203	**B1**	1.083	1.144	2.227	0.061	0.042
						B2	1.120	1.126	2.246	0.006	0.043
						B3	1.100	1.090	2.189	-0.010	0.042
						B4	1.139	1.104	2.244	-0.035	0.043
						B5	1.128	1.106	2.234	-0.022	0.043
C	1.178	1.174	2.353	0.004	0.217	**C1**	0.730	0.691	1.421	-0.040	0.074
						C2	0.737	0.748	1.486	0.011	0.077
						C3	0.746	0.774	1.519	0.028	0.077
D	1.113	1.068	2.181	0.045	0.205	**D1**	1.158	1.067	2.225	-0.091	0.046
						D2	1.083	1.095	2.179	0.012	0.044
						D3	1.072	1.045	2.117	-0.027	0.043
						D4	0.949	0.977	1.926	0.028	0.038
						D5	1.057	1.135	2.192	0.078	0.043
E	0.933	1.125	2.058	-0.192	0.172	**E1**	0.920	0.909	1.830	-0.011	0.035
						E2	0.981	0.979	1.960	-0.002	0.038
						E3	0.893	0.989	1.882	0.097	0.035
						E4	1.025	1.013	2.038	-0.012	0.040
						E5	1.039	0.967	2.006	-0.072	0.040